Graphs

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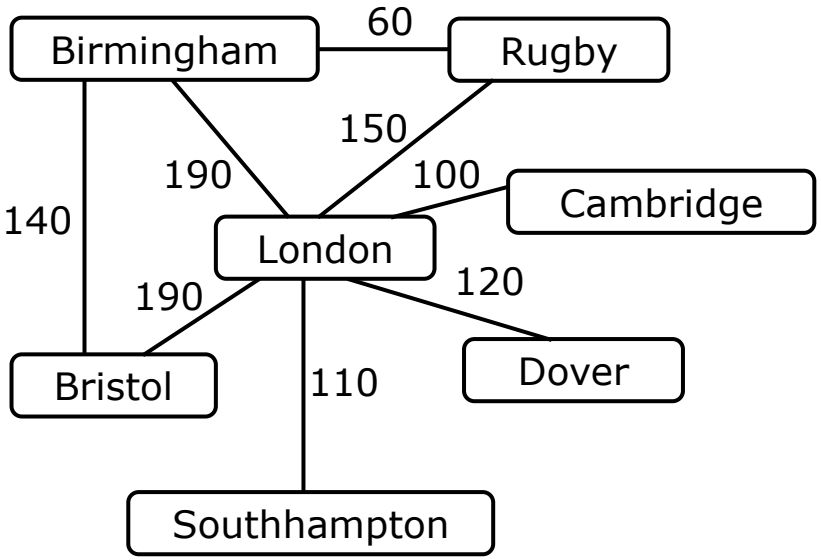
What is a Graph?

- A data structure that consists of a **set of nodes (*vertices*)** and a **set of edges** that relate the nodes to each other
- The set of edges describes relationships among the vertices



Graph

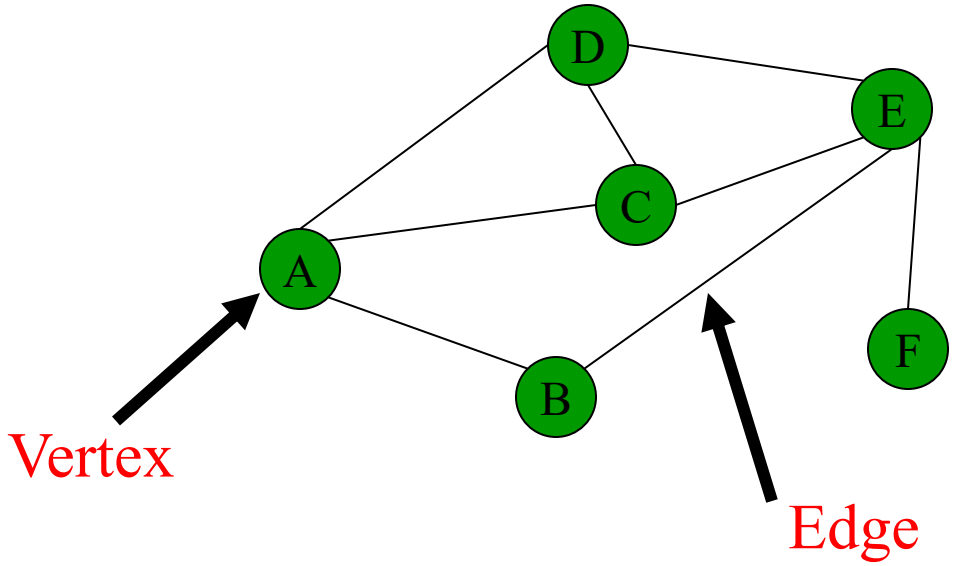
Formal definition of graphs

- A graph *G* is defined as follows:

$$G=(V,E)$$

V(G): a finite, nonempty set of vertices

E(G): a set of edges (pairs of vertices)

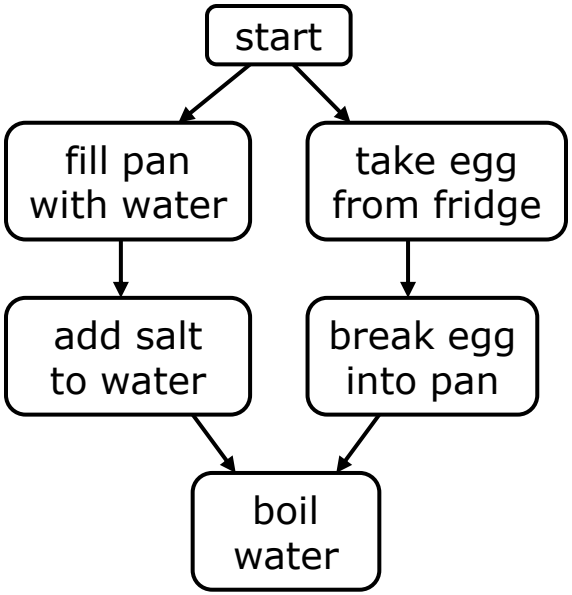


Vertices can be considered "sites" or locations.

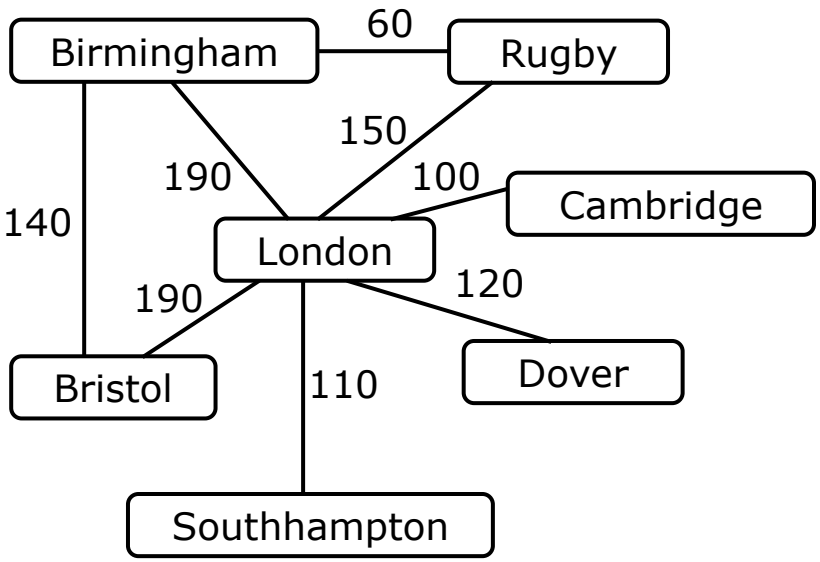
Edges represent connections.

Types of Graph

- There are two kinds of graphs: directed graphs (sometimes called digraphs) and undirected graphs



A directed graph



An undirected graph

Graph terminology I

- A graph is a collection of nodes (or vertices, singular is vertex) and edges (or arcs)
  - Each node contains an element
  - Each edge connects two nodes together (or possibly the same node to itself) and may contain an edge attribute
- A directed graph is one in which the edges have a direction
- An undirected graph is one in which the edges do not have a direction
  - Note: Whether a graph is directed or undirected is a *logical* distinction—it describes how we think about the graph

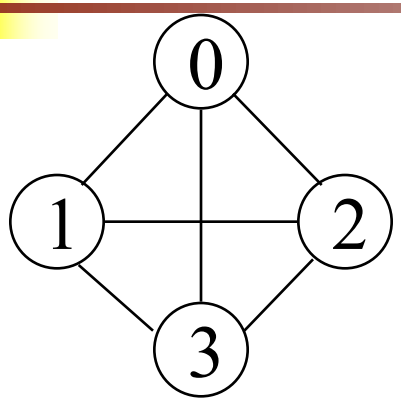
Definition: Directed graph

- Directed Graph
  - A directed graph, or digraph, is a pair
  - G = (V, E)
  - where V is a set whose elements are called vertices, and
  - E is a set of ordered pairs of elements of V.
    - Vertices are often also called nodes.
    - Elements of E are called edges, or directed edges, or arcs.
    - For directed edge (v, w) in E, v is its tail and w its head;
    - (v, w) is represented in the diagrams as the arrow, v -> w.
    - In text we simple write vw.

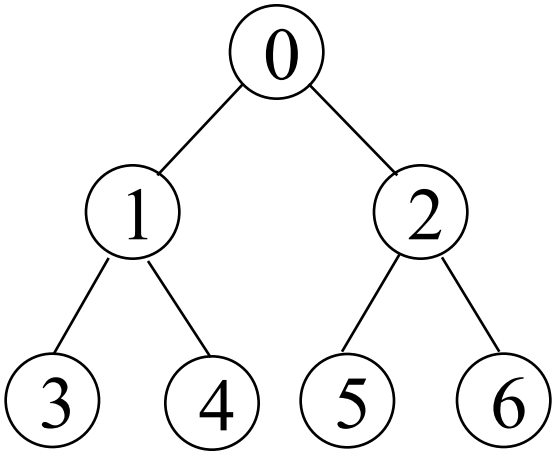
Definition: Undirected graph

- Undirected Graph
  - A undirected graph is a pair
  - G = (V, E)
  - where V is a set whose elements are called vertices, and
  - E is a set of *unordered* pairs of *distinct* elements of V.
    - {v, w} denotes an undirected edge
    - In diagrams this edge is the line v---w.
    - In text we simple write vw, or wv
    - vw is said to be *incident* upon the vertices v and w

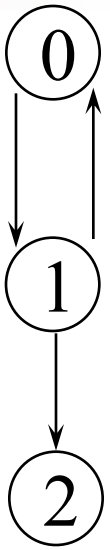
Examples for Graph



G_1
complete graph
Also Cyclic



G_2



incomplete graph

G_3

$V(G_1)=\{0,1,2,3\}$
$V(G_2)=\{0,1,2,3,4,5,6\}$
$V(G_3)=\{0,1,2\}$

$E(G_1)=\{(0,1),(0,2),(0,3),(1,2),(1,3),(2,3)\}$
$E(G_2)=\{(0,1),(0,2),(1,3),(1,4),(2,5),(2,6)\}$
$E(G_3)=\{<0,1>,<1,0>,<1,2>\}$

complete undirected graph: n(n-1)/2 edges
complete directed graph: n(n-1) edges

Complete Graph

- A complete graph is a graph that has the maximum number of edges & all nodes will be connected to each other
  - for undirected graph with n vertices, the maximum number of edges is $n(n-1)/2$
  - for directed graph with n vertices, the maximum number of edges is $n(n-1)$
  - example: G1 is a complete graph